BlueArkive — Product Brief

> ***Product Name***: *BlueArkive*
> ***Tagline***: *Your meetings remember everything. You don't have to.*
> ***Version***: *0.3.6 (Private Beta)*
> ***Platform***: *macOS (primary) · Windows · Linux*
> ***Category***: *AI Meeting Intelligence · Local-First Productivity · Cognitive Memory Fabric*
> ***Website***: *bluearkive.com*
> ***Brief Owner***: *Piyush Kumar — Founder & CEO*
> ***Date***: *May 30, 2026*
> ***Status***: *🟡 Private Beta → Public Beta (Q3 2026)*

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1. Problem Statement

What problem are we solving?

Professionals lose 80% of meeting value within 24 hours. The average knowledge worker attends 15.5 meetings per week (Microsoft Work Trend Index 2025), yet retains less than 20% of what was discussed by the next day (Ebbinghaus forgetting curve). This creates a compounding organizational amnesia:

- **Action items vanish.** 63% of action items from meetings are never completed because they weren't captured or were lost in scattered notes.
- **Decisions are forgotten.** Teams re-discuss the same topics across multiple meetings, wasting an estimated 31 hours/month per employee.
- **Context collapses.** When a team member asks "what did we decide about X three weeks ago?", no one can answer with certainty. The institutional memory lives in someone's head — or nowhere.

Why existing solutions fail

Pain Point	Current State	Why It Fails
Privacy	Existing tools send all audio to cloud servers	Regulated industries (healthcare, legal, finance) cannot use them. HIPAA/SOC2 violations.
Intelligence	Most tools are transcript dumps — keyword search at best	No understanding of *relationships* between meetings. No contradiction detection. No temporal reasoning.
Ownership	Data lives on vendor servers	Vendor lock-in. If a provider shuts down or raises prices, your entire meeting history is hostage.
Integration	Standalone tools that don't connect to workflow	Transcripts sit in a silo. No knowledge graph. No cross-meeting action item tracking.
Cost	$20-40/user/month for basic transcription	Unsustainable for teams of 10+. Most of the cost pays for cloud compute, not intelligence.

The deeper insight

The real problem isn't transcription — it's **cognitive decay**. Meetings generate the richest, most context-dense knowledge in an organization, yet it's the *least structured and least retrievable*. Email has search. Code has Git. Documents have version history. But meetings? They evaporate.

BlueArkive treats meetings as first-class knowledge objects — transcribed, encrypted, semantically indexed, connected through a knowledge graph, and queryable with natural language — all without your data ever leaving your device.

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2. Product Vision

> ***We are building the world's first sovereign cognitive memory fabric for meetings.***

BlueArkive is not a transcription tool. It's a **local-first intelligence layer** that transforms ephemeral conversations into persistent, queryable, interconnected organizational memory — with the privacy guarantees that regulated industries demand and the cognitive depth that cloud-first architectures fundamentally cannot deliver.

Vision Statement (3-Year)

By 2029, BlueArkive will be the default meeting intelligence platform for privacy-conscious organizations — from solo consultants to Fortune 500 legal departments — replacing the fragile cloud-dependent tools of the 2020s with a sovereign, AI-native knowledge fabric that users actually own.

Core Design Principles

Principle	What It Means	How We Enforce It
Local-First	Core functionality works without internet. Your data lives on your device.	All AI inference, transcription, search, and encryption run via edge-optimized machine learning and embedded local storage.
Privacy by Architecture	Privacy isn't a feature toggle — it's a structural guarantee.	Military-grade AES-256-GCM encryption with rigorous key derivation. Keys stored securely in the native OS keychain. Zero-knowledge cloud sync.
Cognitive Depth	Go beyond transcription to understanding.	Knowledge graph with entity extraction, contradiction detection, sentiment analysis, and temporal reasoning.
Data Sovereignty	Users own their data. Period.	24-word BIP39 recovery phrase. Crypto-shredding capability. Self-hostable backend. No vendor lock-in.
Hardware-Adaptive	Performance should match the device.	3-tier hardware detection (high/mid/low) with Apple Silicon GPU acceleration and battery-aware scheduling.

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3. Product Objectives

Primary Objectives (v1.0 — Public Launch)

#	Objective	Key Result	Target Date

O1	**Ship production-grade local transcription**	< 3s latency on Apple Silicon, 95%+ WER accuracy via Whisper	Q3 2026
O2	**Deliver "Ask Your Meetings" intelligence**	Users can query across all meetings with natural language and get cited answers	Q3 2026
O3	**Achieve enterprise-grade security certification**	Pass SOC2 Type I audit; HIPAA-compliant architecture validated	Q4 2026
O4	**Establish product-market fit signal**	500 active beta users, 40%+ weekly retention, NPS > 50	Q4 2026
O5	**Launch pricing and billing**	Functional 5-tier billing (Free → Enterprise) with Razorpay + LemonSqueezy	Q3 2026

Stretch Objectives (v1.5+)

#	Objective	Key Result	Target Date
S1	**Team collaboration**	Shared workspaces with role-based access, centralized billing	Q1 2027
S2	**Mobile companion app**	iOS/Android app for meeting playback and search (read-only MVP)	Q2 2027
S3	**Calendar-aware auto-recording**	Auto-start transcription when a calendar meeting begins	Q1 2027
S4	**Webhook ecosystem**	Event-driven integrations (Slack, Notion, Linear, Jira) via webhooks	Q1 2027

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4. Target Audience

Primary Personas

Persona 1: "The Solo Consultant" — Maya, 34

Attribute	Detail
Role	Independent management consultant
Pain	Takes 8-12 client meetings/week. Spends 2+ hours/day writing up notes. Can't search across past client conversations.
Current Tools	Apple Notes + Voice Memos + manual transcription
Why BlueArkive	Automatic transcription saves 10+ hrs/week. Cross-meeting search means she can recall what Client X said about budget 3 months ago in seconds. Client confidentiality demands local-first.
Tier	Pro ($19/mo)

Deal Breaker	If any client audio touches a third-party server

Persona 2: "The Engineering Lead" — Raj, 29

Attribute	Detail
Role	Senior Engineering Manager at a 200-person startup
Pain	15+ meetings/week. Decisions made in standups get lost. Action items from sprint retros are never tracked. New hires have zero context on past architectural decisions.
Current Tools	Google Meet recordings (which nobody re-watches) + Notion (manually updated)
Why BlueArkive	Knowledge graph connects decisions across sprints. "Ask Meetings" gives new hires instant context. Weekly digest auto-summarizes what happened.
Tier	Team ($15/user/mo)
Deal Breaker	If it requires admin overhead to set up or manage

Persona 3: "The Compliance Officer" — Dr. Sarah, 42

Attribute	Detail
Role	Chief Compliance Officer at a regional hospital network
Pain	Clinical team meetings discuss PHI. Federal regulations prohibit sending audio to cloud services. Current meeting tools are banned by IT.
Current Tools	Manual note-taking + locked filing cabinets
Why BlueArkive	HIPAA-compliant architecture (local processing + encryption + audit logs). PHI auto-detection prevents accidental data leakage through comprehensive scanning of sensitive identifiers.
Tier	Enterprise (Custom)
Deal Breaker	If it can't pass their security audit

Market Segmentation

Segment	TAM Estimate	Priority	Why
Solo Knowledge Workers (consultants, lawyers, therapists)	~12M users globally	🔴 P0 — Launch segment	Low acquisition cost, high pain, privacy-sensitive, word-of-mouth driven
Startup Engineering Teams (5-50 person)	~800K teams globally	🟠 P1 — Post-launch	Team features drive expansion revenue, strong PLG motion
Regulated Enterprise (healthcare, legal, finance)	~2M organizations globally	🟡 P2 — v1.5+	Highest ARPU, longest sales cycle, requires compliance certifications

5. Key Features & Functionality

5.1 Core Features (v1.0 — Must Ship)

🎙️ Real-Time Local Transcription

- **What**: Live speech-to-text using state-of-the-art transcription models running entirely locally, with simultaneous mic and system audio capture.
- **Why it matters**: Captures both sides of the conversation (you and the remote participants on Zoom/Teams) securely without virtual audio cables. < 3s latency on Apple Silicon.
- **Technical**: Native multi-channel capture for system and microphone audio. Advanced voice activity detection and intelligent chunking ensure high accuracy and performance. Hardware-adaptive model selection.

👥 Local Speaker Diarization

- **What**: Identifies 'Speaker A' vs 'Speaker B' entirely on-device.
- **Why it matters**: A transcript is useless if you don't know who agreed to the deal. Local diarization maintains privacy while structuring the conversation.
- **Technical**: Integrated into the transcription pipeline, avoiding expensive and privacy-violating cloud API calls.

🔐 End-to-End Encryption

- **What**: AES-256-GCM encryption with rigorous key derivation
- **Why it matters**: Even if a device is stolen, data is unreadable without the user's passphrase
- **Technical**: Keys stored securely in the native OS keychain. Standard 24-word recovery phrase. Cryptographic keys are aggressively zeroed from memory on lock or shutdown. Crypto-shredding guarantees all data is permanently unrecoverable upon key deletion.

🔍 Semantic Search & Chat ("Ask Your Meetings")

- **What**: A "ChatGPT-like" conversational interface with real-time token streaming, capable of natural language queries across all historical transcripts and notes.
- **Why it matters**: Keyword search fails when you forget exact phrasing. "What did we decide about the pricing model?" returns the exact moment with synthesized answers and interactive source citations.
- **Technical**: Utilizes an advanced **Cognitive RAG Pipeline** with a **3-tier search escalation** architecture: (1) Instant local full-text search, (2) Local semantic search via embedded models, and (3) Cloud-hybrid semantic and fuzzy search (tier-gated). Results are rank-fused across tiers. Conversational state is managed natively with streaming token support and mid-generation cancellation, freeing hardware resources immediately. Crash-safe local indexing ensures query availability immediately post-meeting.

🧠 Bitemporal Knowledge Graph

- **What**: Automatically extracts entities (people, topics, decisions, action items) and maps multi-hop relationships across your entire meeting history.
- **Why it matters**: Transcripts are static; a knowledge graph is a living memory. It surfaces connections humans miss — e.g., "Every time we discuss pricing, Sarah raises compliance concerns."
- **Technical**: Advanced entity extraction via local machine learning models. Powered by an internal **Conflict Resolution Engine**, utilizing intelligent logic to detect temporal contradictions (e.g., a decision made in Q1 reversed in Q2). Visualized locally via an interactive force-directed graph.

📝 AI-Powered Note Expansion & Derivation

- **What**: A rich text editor (TipTap) that acts as an autonomous scribe. It uses the raw, multi-speaker transcript to auto-expand shorthand notes, summarize themes, and extract assignee-mapped action items.
- **Why it matters**: Turns messy, shorthand meeting notes into structured, shareable PRDs or executive summaries in one click, eliminating post-meeting administrative work.
- **Technical**: Context-aware AI processing grounded in intelligently chunked transcripts. Derives underlying intent rather than just summarizing text. Includes deterministic action item extraction.

💡 **Live AI Coach & Silent Prompter**

- **What**: Real-time, in-meeting suggestions that push directly to the OS overlay based on meeting context, supporting **4 prompt modes**: title suggestion, follow-up question, action item extraction, and decision summarization.
- **Why it matters**: Guides negotiations, reminds you to ask qualifying questions, or notes when you are dominating the conversation without distracting you. Each mode produces a structurally different output — not just generic summaries.
- **Technical**: Contextually generates suggestions from recent transcript history. **Dual-path execution**: Premium cloud execution for advanced modes, with a local on-device fallback always available. Model parameters adapt dynamically per mode to optimize for analytical precision or creative questioning. Seamlessly updates the OS widget state.

📅 **Automated Weekly Digest**

- **What**: A synthesized, cross-meeting report generated at the end of the week, summarizing key decisions, persistent blockers, and macro-trends across all conversations.
- **Why it matters**: Gives executives and managers a "bird's-eye view" of their week without re-reading transcripts. Highlights unresolved action items automatically.
- **Technical**: Aggregates local meeting statistics and fuses them with AI-derived macro-insights. Works gracefully offline by prioritizing local heuristics, falling back to cloud enrichment only if permitted by tier.

🏝️ **Persistent OS Capture (Dynamic Island) & Global Shortcuts**

- **What**: A floating, always-on top OS-level widget (Dynamic Island style) that provides spatial handoff and **full meeting control without restoring the main window** — including quick notes, bookmarks, pause/resume, and start capture — paired with global keyboard shortcuts.
- **Why it matters**: You shouldn't have to hunt for a window to start recording or view a coach tip. The widget is a complete control surface: submit a quick note, drop a bookmark on an important moment, or pause recording — all without switching apps.
- **Technical**: Native-feeling OS overlay with robust window detection. Secure proxy channels for quick actions with smooth spatial handoff animations. Visibility is auto-managed based on recording state, paired with system-wide hotkeys.

🎓 **Interactive Ghost Meeting Tutorial**

- **What**: A state-machine-driven onboarding experience that simulates a real meeting — live transcript streaming, AI Coach suggestions, note expansion, and Dynamic Island takeover — before the user's first real session.
- **Why it matters**: Most productivity tools overwhelm new users with feature lists. BlueArkive *shows* the product in action through a 5-step interactive tutorial, building muscle memory before the first real meeting. Users experience the "aha moment" in 90 seconds, not 90 minutes.
- **Technical**: 5-phase state machine driving an interactive tutorial. Uses real UI components — not mockups — for live transcript streaming, AI Coach demonstrations, and note expansion. Features typewriter animations for AI responses and a progress indicator, with the ability to skip anytime.

🩺 **Self-Healing Health Dashboard**

- **What**: A 7-system diagnostic panel that tests all critical subsystems (Database, Auth, Microphone, Screen Recording, Network, Disk Space, Native Modules) and offers one-click proactive repair actions.
- **Why it matters**: Enterprise users and IT admins need to verify system health without contacting support. The dashboard turns "it's not working" support tickets into self-service fixes.
- **Technical**: Runs 7 diagnostic probes in sequence covering database connectivity, authentication, native OS media access, network health, disk space, and core modules. Provides direct repair actions such as requesting OS permissions, deep-linking to system settings, or self-service search index rebuilding.

📱 Device Management

- **What**: Multi-device registration, tier-gated device limits, remote deactivation, and device renaming — with structured error responses when limits are exceeded.
- **Why it matters**: Users need to manage which devices have access to their meeting data. Tier-gated device limits drive upgrades (Free: 1 device, Starter: 2, Pro+: unlimited).
- **Technical**: Centralized device management service supporting registration and limits based on subscription tier. Limit-exceeded scenarios are handled gracefully to present contextual upgrade paths.

🔋 Battery-Aware Resource Scheduling

- **What**: Automatic detection of battery vs. AC power state, with resource reduction when on battery to preserve laptop battery during long meetings.
- **Why it matters**: A 2-hour meeting on battery shouldn't kill your laptop. BlueArkive automatically reduces AI inference frequency, defers background embedding, and throttles sync when unplugged.
- **Technical**: Wraps native OS power monitoring APIs. Battery state is propagated throughout the application, automatically instructing resource-intensive subsystems to scale down when unplugged.

🔄 Encrypted Cloud Sync

- **What**: Multi-device sync with end-to-end encryption. Data is encrypted before it leaves the device.
- **Why it matters**: Use BlueArkive on your MacBook at work and your iMac at home without compromising privacy
- **Technical**: Vector clock-based conflict resolution and CRDT compatibility ensure robust multi-device sync. Data is synced atomically with strict schema whitelisting.

5.2 Differentiating Features

Feature	Capability	Industry Standard
Local-first processing	✅ All AI runs locally on-device	Most tools require cloud processing
System Audio Capture	✅ WASAPI / ScreenCaptureKit native	Often requires clunky 3rd party drivers
Local Speaker Diarization	✅ Identifies speakers on-device	Almost exclusively cloud-based
End-to-end encryption	✅ AES-256-GCM + BIP39 recovery	Rare in meeting tools
Cross-meeting knowledge graph	✅ Entity + relationship mapping	Not available elsewhere

Contradiction detection	✅ Temporal reasoning across meetings	Not available elsewhere
PHI auto-detection	✅ 17 identifiers + Luhn validation	Not available elsewhere
Live AI Coach	✅ Real-time contextual nudges	Not available elsewhere
Dynamic Island & Hotkeys	✅ Persistent spatial handoff UI	Not available elsewhere
Hardware-adaptive AI	✅ 3-tier + GPU + battery-aware	Not available elsewhere
Self-hostable backend	✅ IBackendProvider abstraction	Rare in meeting tools
Crypto-shredding	✅ Delete keys = delete all data	Not available elsewhere
Weekly meeting digest	✅ Auto-generated summaries	Limited availability
Offline functionality	✅ Full (transcription + search + notes)	Rare — most require internet
Sentiment analysis	✅ Per-segment emotional tone	Limited availability
Audit logs (HIPAA)	✅ Tamper-evident logging + CSV export	Not available elsewhere
Ghost Meeting onboarding	✅ Interactive tutorial with real UI components	Not available elsewhere
Self-healing diagnostics	✅ 7-system health check + 1-click repair	Not available elsewhere
Multi-mode AI suggestions	✅ 4 modes (title/question/action/decision)	Not available elsewhere
Cancellable AI generation	✅ AbortController mid-stream cancellation	Not available elsewhere
Widget quick actions	✅ Notes/bookmarks/pause from widget	Not available elsewhere
Battery-aware scheduling	✅ Auto-reduces resources on battery	Rare in meeting tools
3-tier search escalation	✅ FTS5 → Semantic → Cloud Hybrid	Not available elsewhere
Device management	✅ Multi-device registration + tier limits	Limited availability

5.3 Feature Tiers (Pricing Gates)

Feature	Free	Starter ($9/mo)	Pro ($19/mo)	Team ($15/user/mo)	Enterprise
Local transcription	✅	✅	✅	✅	✅
Transcript limit	5K chars	15K chars	50K chars	100K chars	100K chars

Cloud sync	—	✅	✅	✅	✅
Devices	1	2	Unlimited	Unlimited	Unlimited
AI queries/mo	—	50	Unlimited	Unlimited	Unlimited
Knowledge Graph	View only	View only	Interactive	Interactive	Interactive
Speaker diarization	—	✅	✅	✅	✅
Weekly digest	—	✅	✅	✅	✅
Hybrid search	—	—	✅	✅	✅
Calendar sync	—	✅	✅	✅	✅
Calendar auto-link	—	—	✅	✅	✅
Webhooks	—	3	10	Unlimited	Unlimited
Team collaboration	—	—	—	✅	✅
Audit logs	—	—	—	—	✅
SSO / SAML	—	—	—	—	✅
Custom SLA	—	—	—	—	✅
Sentiment analysis	✅	✅	✅	✅	✅
Action items	✅	✅	✅	✅	✅
Dynamic Island Widget	✅	✅	✅	✅	✅
Widget quick actions	✅	✅	✅	✅	✅
Health Dashboard	✅	✅	✅	✅	✅
Ghost Meeting tutorial	✅	✅	✅	✅	✅
Device management	1 device	2 devices	Unlimited	Unlimited	Unlimited
Live AI Coach	—	—	✅	✅	✅
Multi-mode suggestions	—	Title only	All 4 modes	All 4 modes	All 4 modes

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6. Strategic Advantages

Why BlueArkive Wins

1. **Privacy is structural, not promised.** BlueArkive *architecturally cannot* access your data — keys live in your OS keychain, not on our servers. This is a design guarantee, not a policy promise.
2. **Intelligence compounds over time.** A transcript is a snapshot. A knowledge graph is a living memory. BlueArkive gets smarter with every meeting — after 50 meetings, the intelligence gap is uncatchable.
3. **Regulated industries have no alternative.** A HIPAA-compliant meeting intelligence tool that works offline with E2E encryption and PHI detection doesn't exist today. We are creating the category.
4. **Cloud economics favor us.** Cloud-based transcription costs ~$0.006/minute per user. Our marginal cost per user is near zero — inference runs on the user's own hardware. This enables a structurally lower price point with higher margins.

Latency: BlueArkive vs Cluely

Operation	BlueArkive (Local)	Cluely (Cloud)	Winner
STT Processing	0.58s per 30s audio (Whisper Turbo)	Network RTT + Deepgram API queue	🏆 BlueArkive
STT (low-tier)	34ms per 10s audio (Moonshine)	Same cloud dependency	🏆 BlueArkive
LLM First Token	~130ms (Qwen 2.5 3B local)	5–90 seconds (reported by users/Business Insider)	🏆 BlueArkive
Search (FTS5)	<1ms average across 100K segments	Pinecone API call + network	🏆 BlueArkive
Database Writes	75,188 inserts/sec (SQLite WAL)	Neon serverless cold-start + network	🏆 BlueArkive
Offline Latency	Same as above — zero degradation	∞ — completely non-functional	🏆 BlueArkive
VAD Processing	<10ms per chunk (Silero ONNX)	Unknown (likely server-side)	🏆 BlueArkive

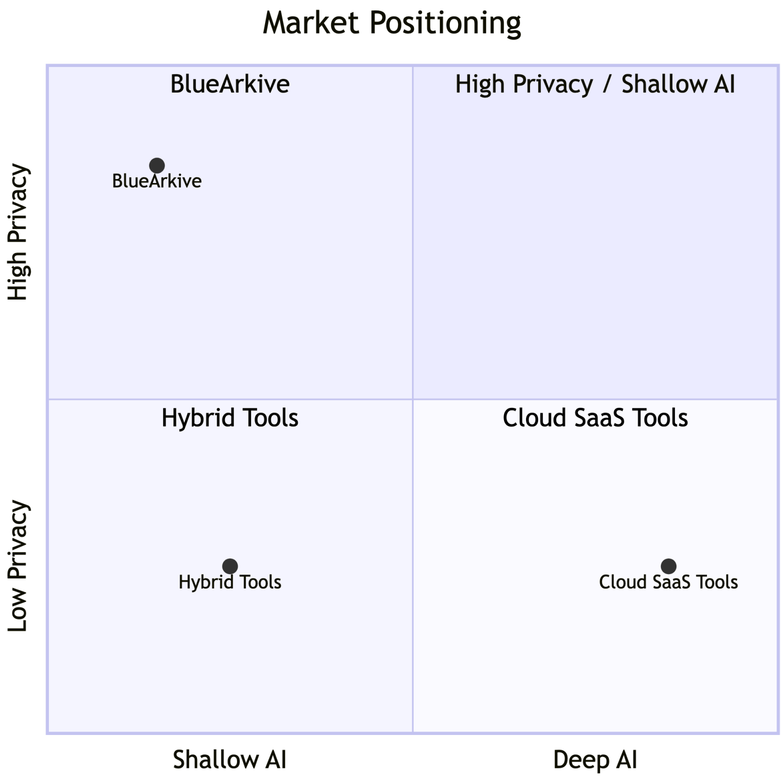
Why the gap is so massive

- **BlueArkive's latency** = compute time only — no network hop, no API queue, no cold start.
- **Cluely's latency** = network RTT + API queue + vendor processing + return trip — for every single request, through multiple vendors in series (Deepgram → Pinecone → OpenAI → Pusher).

The 5–90 second latency range reported for Cluely makes it essentially useless for real-time interview/meeting assistance — the very thing it's marketed for. BlueArkive's 130ms time-to-first-token is **~40-700× faster**.

BlueArkive wins latency by 1-3 orders of magnitude. 🏆

Market Positioning



BlueArkive occupies the **top-left quadrant** — deep AI with high privacy — a position that is structurally difficult for cloud-first incumbents to reach because privacy requires architectural commitment, not feature additions.

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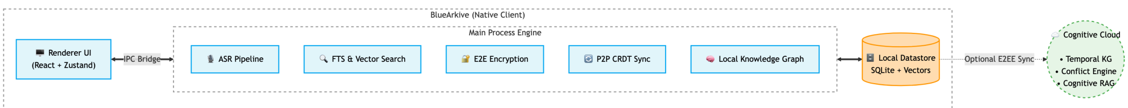
7. Technical Architecture

System Overview



Technology Stack

Layer	Technology	Why
Runtime	Modern secure native wrapper	Cross-platform native app with system-level audio access
Frontend	Performant Web Technologies	Smooth state management, declarative UI
Database	Crash-safe embedded database	< 1ms reads, fully local, enterprise-ready datastore

Search	Embedded Search (FTS + Vector)	Hybrid keyword + semantic search without cloud dependency
ASR	State-of-the-art ASR	High accuracy, runs locally with hardware acceleration
Encryption	AES-256-GCM + PBKDF2	Military-grade encryption with OS-native key storage
Visualization	Native interactive data mapping	Interactive knowledge graph with dynamic capability
Sync	Vector clocks + CRDT	Conflict-free multi-device sync without central authority
Cloud Backend	Cognitive Cloud Backend	Advanced cognitive engine with bitemporal KG and tool integrations
Billing	Razorpay (India) + LemonSqueezy (Global)	Dual payment gateway for global + India-specific pricing (INR)

Service Architecture (40 Services)

Category	Services	Purpose
Audio	ASRService, AudioPipelineService, AudioCacheCleanup	Capture → VAD → Whisper → Transcript
Intelligence	LocalEntityExtractor, SentimentAnalyzer, ActionItemService	Entity extraction, sentiment, action items
Storage	DatabaseService, MigrationService, BackgroundEmbeddingQueue	SQLite ops, schema migrations, async embedding
Search	LocalEmbeddingService (FTS5 + vectors)	Hybrid semantic + keyword search
Security	EncryptionService, KeyStorageService, RecoveryPhraseService, PHIDetectionService, AuditLogger	E2E encryption, key management, PHI scanning, audit trail
Sync	SyncManager, ConflictResolver, VectorClockManager, YjsConflictResolver, DeviceManager	Multi-device sync with CRDT conflict resolution
Cloud	AuthService, CloudAccessManager, CloudTranscriptionService	Authentication, tier gating, cloud AI fallback
Backend	Multi-Backend Provider Abstraction	Provider registry pattern for seamless backend swapping
Platform	Hardware & Model Lifecycle Services	Hardware detection, pricing tiers, model lifecycle

Ops	Observability & Diagnostic Services	Observability, crash reporting, rate limiting, webhooks
DI	Dependency Injection Container	Lazy singleton dependency injection for core services

Security Architecture (13 Controls)

#	Control	Implementation
1	**AES-256-GCM Encryption**	All data encrypted at rest with rigorously derived keys
2	**OS Keychain Storage**	Keys stored natively — never on disk or in source code
3	**BIP39 Recovery**	24-word mnemonic for user-controlled key recovery
4	**Crypto-Shredding**	Delete keys → all synced data permanently unrecoverable
5	**Key Zeroing**	Cryptographic keys aggressively wiped from memory on lock/logout/shutdown
6	**Context Isolation**	Strict sandboxing and disabled node integration
7	**CSP Header**	Content-Security-Policy blocks unauthorized code execution
8	**Navigation Guard**	Navigation handlers prevent renderer hijacking
9	**Injection Prevention**	Strict schema whitelisting for all data synchronization operations
10	**Index Crash Safety**	Per-table error isolation ensures no search outage
11	**PHI Detection**	Comprehensive detection of sensitive identifiers
12	**Audit Logging**	Tamper-evident logs for HIPAA compliance
13	**Web Security**	Strict web security enforced on all application windows

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8. User Stories

Epic 1: Meeting Capture

ID	As a...	I want to...	So that...	Priority	Status
US-101	Knowledge worker	Start recording a meeting with one click	I don't have to manually take notes	P0	✅ Done
US-102	Consultant	See live transcription as people speak	I can follow along and correct errors in real-time	P0	✅ Done

US-103	Privacy-conscious user	Have all transcription happen on my device	No audio ever leaves my machine	P0	✅ Done
US-104	User with limited hardware	Have the app adapt AI quality to my hardware	It doesn't crash or drain my battery on older machines	P1	✅ Done
US-105	Mobile professional	Record meetings offline (airplane, poor WiFi)	I never miss capturing a conversation	P0	✅ Done

Epic 2: Intelligence & Search

ID	As a...	I want to...	So that...	Priority	Status
US-201	Engineering lead	Ask "What did we decide about the database migration?"	I get the exact answer with the meeting source cited	P0	✅ Done
US-202	Consultant	See all meetings where "Client X" was discussed	I can prepare for a follow-up meeting in 2 minutes	P0	✅ Done
US-203	Product manager	View a knowledge graph of how topics connect	I can see patterns across a quarter of meetings	P1	✅ Done
US-204	Team lead	Get a weekly digest of all meetings	I can catch up on what my team discussed without watching recordings	P1	✅ Done
US-205	Compliance officer	Detect when a meeting conclusion contradicts a prior decision	I can flag inconsistencies before they become problems	P2	🟡 Beta

Epic 3: Security & Privacy

ID	As a...	I want to...	So that...	Priority	Status
US-301	Healthcare professional	Have PHI automatically detected before any sync	Patient information never leaves my device	P0	✅ Done
US-302	Enterprise admin	View tamper-evident audit logs	I can prove compliance during SOC2 audits	P1	✅ Done
US-303	Departing employee	Crypto-shred all my data with one action	My meeting data is permanently unrecoverable after I leave	P0	✅ Done

US-304	Multi-device user	Sync meetings across devices with E2E encryption	I can access meetings on my work laptop and home desktop	P0	✅ Done

Epic 4: Team Collaboration

ID	As a...	I want to...	So that...	Priority	Status
US-401	Team admin	Create a shared workspace for my team	We can pool meeting intelligence across the team	P2	🔜 Planned
US-402	Manager	Get webhook notifications when action items are created	They flow into our project tracker (Linear, Jira) automatically	P2	🟡 Beta
US-403	New hire	Search all team meeting history for onboarding context	I can ramp up in days instead of weeks	P2	🔜 Planned

Epic 5: Onboarding & Self-Service

ID	As a...	I want to...	So that...	Priority	Status
US-501	First-time user	Experience a simulated meeting tutorial before my first real recording	I understand the product's value in 90 seconds without reading docs	P0	✅ Done
US-502	Non-technical user	See a health dashboard showing if mic/network/database are working	I can self-diagnose issues without contacting support	P1	✅ Done
US-503	User with broken search	Rebuild my search indexes from the health dashboard	I can fix corrupted FTS5 indexes without reinstalling the app	P1	✅ Done
US-504	Multi-device user	See which devices are registered and deactivate old ones	I can manage my device limit and free slots for new machines	P1	✅ Done
US-505	Laptop user on battery	Have the app automatically reduce resource usage on battery	My laptop battery isn't drained during a long meeting	P1	✅ Done

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9. Timeline & Milestones

Phase 1: Foundation (Completed — Q4 2025 – Q2 2026)

Milestone	Status	Deliverables
Core transcription engine	✅ Complete	State-of-the-art speech models, voice activity pipeline, real-time streaming
Encryption infrastructure	✅ Complete	AES-256-GCM, native OS keychain integration, standard recovery
Database layer	✅ Complete	Embedded datastore, local search indexing, vector embeddings
UI shell	✅ Complete	Meeting list, detail view, notes editor, settings, onboarding
Backend abstraction	✅ Complete	Robust multi-provider abstraction and strict factory patterns
Security hardening	✅ Complete	Application sandboxing, comprehensive PHI detection, secure memory handling
Knowledge graph MVP	✅ Complete	Entity extraction, interactive visualization, basic relationship mapping

Phase 2: Intelligence & Polish (Current — Q3 2026)

Milestone	Target	Deliverables
Ask Meetings v1	July 2026	Semantic search with cited answers, context-aware responses
Weekly Digest v1	July 2026	Auto-generated meeting summaries, trend detection
Speaker Diarization	Aug 2026	Multi-speaker identification and labeling
Pricing & Billing launch	Aug 2026	Razorpay + LemonSqueezy integration, tier enforcement
Public Beta launch	Sep 2026	Marketing site update, ProductHunt launch, beta waitlist conversion

Phase 3: Growth & Enterprise (Q4 2026 – Q1 2027)

Milestone	Target	Deliverables
SOC2 Type I certification	Nov 2026	Security audit, compliance documentation
Team workspaces v1	Dec 2026	Shared workspaces, admin controls, role-based access
Calendar integration	Jan 2027	Google Calendar + Outlook sync, auto-recording
Webhook ecosystem v1	Jan 2027	Slack, Notion, Linear, Jira integrations
Enterprise pilot program	Feb 2027	Custom SLA, SSO/SAML, dedicated support

Phase 4: Scale (Q2 2027+)

Milestone	Target	Deliverables
Mobile companion app	Q2 2027	iOS/Android read-only app for search and playback
Windows/Linux stable	Q2 2027	Parity with macOS feature set
Self-hosted enterprise	Q3 2027	On-premises Cognitive Engine deployment package
API & developer platform	Q3 2027	Public API for custom integrations

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10. Success Metrics

North Star Metric

> ***Weekly Active Meeting Hours Indexed (WAMHI)*** *— Total hours of meeting audio transcribed and indexed across all users per week.*

Why this metric: It captures both user acquisition (more users) and engagement depth (more meetings per user). A user who records one meeting is experimenting. A user who records 10 meetings/week has made BlueArkive essential.

Leading Indicators

Metric	Definition	Target (Month 3)	Target (Month 6)	Target (Month 12)
WAU (Weekly Active Users)	Unique users who open the app	200	1,000	5,000
Meetings/User/Week	Avg meeting recordings per active user	2.5	4.0	6.0
Search Queries/User/Week	"Ask Meetings" queries per active user	1.0	3.0	5.0
D7 Retention	% of new users active 7 days after signup	35%	45%	55%
D30 Retention	% of new users active 30 days after signup	15%	25%	35%
NPS	Net Promoter Score (quarterly survey)	40	50	60+
Free → Paid Conversion	% of free users upgrading within 30 days	3%	5%	8%

Revenue Metrics

Metric	Target (Month 6)	Target (Month 12)

MRR (Monthly Recurring Revenue)	$5,000	$30,000
ARPU (Avg Revenue Per User)	$12	$15
Churn Rate (Monthly)	< 8%	< 5%
LTV:CAC Ratio	> 2:1	> 3:1

Technical Health Metrics

Metric	Target
Transcription accuracy (WER)	> 95% on native English audio
App crash rate	< 0.5% of sessions
Search latency (p95)	< 500ms for hybrid semantic and keyword
Cold start time	< 4s on modern silicon, < 8s on legacy
Database size efficiency	< 2MB per hour of meeting audio indexed
Codebase stability	0 errors across strict type-checking

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11. Budget & Resources

Infrastructure Costs (Monthly)

Line Item	Cost	Notes
Cloud Infrastructure (Database + Vector Storage)	$150/mo	Enterprise-tier hosting
Domain + DNS (bluearkive.com)	$15/mo	Cloudflare
Vercel Hosting (landing page)	$0-20/mo	Pro plan
Sentry (crash reporting)	$0/mo	Free tier
Apple Developer Program	$8.25/mo	$99/year for macOS signing
Code signing (Windows)	~$30/mo	DigiCert EV certificate (amortized)
Total	**~$225/mo**	

Cost Advantages

BlueArkive	Cloud-Based Alternatives
$0.00/user for transcription (runs on user hardware)	$0.006/min × 900 min/mo = $5.40/user/mo
$0.00/user for embedding (local inference)	$0.02/1K tokens × ~50K tokens/mo = $1.00/user/mo
$0.00/user for storage (local datastore)	$0.023/GB × ~2GB/user = $0.05/user/mo

Marginal cost per user: ~$0	**Marginal cost per user: ~$6.45/mo**

> *At 10,000 users, cloud-based services spend ~$64,500/month on compute. BlueArkive spends ~$225/month total. This is a **structural cost advantage** that allows us to offer a better product at a lower price while maintaining higher margins.*

Funding Requirements

Phase	Amount	Use
Pre-Seed (Current)	Bootstrapped	Product development, initial users
Seed (Target: Q4 2026)	$500K – $1M	2 engineers, go-to-market, SOC2 certification, first 1,000 users
Series A (Target: Q3 2027)	$3M – $5M	Enterprise sales team, mobile apps, international expansion

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12. Risks & Mitigations

#	Risk	Likelihood	Impact	Mitigation
R1	**Whisper accuracy insufficient for accented English / non-English**	Medium	High	Support model swapping. Offer cloud transcription as premium fallback (already built: `CloudTranscriptionService`). Monitor WER by language.
R2	**Apple blocks Electron apps or restricts audio capture**	Low	Critical	Build native macOS Swift UI fallback. Monitor Apple developer announcements. Maintain relationship with Apple developer relations.
R3	**Cloud incumbent adds local processing mode**	Medium	High	Cloud business models require server-side data. Our moat is the *knowledge graph* + encryption, not just local processing.
R4	**Enterprise sales cycle too long for bootstrapped startup**	High	Medium	Focus on PLG (Product-Led Growth) with solo users first. Enterprise comes after product-market fit with individuals.
R5	**Large model downloads deter first-time users**	Medium	Medium	Progressive download with `ModelDownloadService`. Start with distilled model (40MB). Full model downloads in background.
R6	**SQLite performance degrades at scale (1000+ meetings)**	Low	Medium	WAL mode + async embedding queue already handle this. If needed, migrate to local PostgreSQL via IBackendProvider abstraction.
R7	**Cloud incumbent copies our privacy**	Medium	Low	Privacy requires *architectural* commitment, not marketing. Retrofitting E2E encryption

	positioning			into a cloud-first product is a multi-year project. First-mover advantage is real.
R8	**Regulation (GDPR/CCPA) creates compliance burden**	Low	Medium	Local-first architecture is *inherently* compliant — we don't process or store user data. Crypto-shredding satisfies right-to-deletion.

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14. Appendix

A. Codebase Snapshot (May 2026)

Metric	Value
Client Architecture	Modular, strictly typed
Backend Architecture	Proprietary Cognitive Engine
Backend testing	Extensive comprehensive suite
Security Architecture	Native sandboxing + isolation
App version	0.3.6

B. Key Dependencies

Infrastructure Layer	Purpose
Embedded Database	High-performance local datastore
Inference Engine	Local AI execution and processing
Keychain Manager	OS keychain access for secure storage
Data Visualization	Knowledge graph mapping and rendering
State Management	Performant application state
Editor Framework	Rich text notes and structured documents
Telemetry	Crash reporting and diagnostics
Distribution	Seamless auto-updates

C. Cognitive Engine (Cloud Backend)

Module	Function
Temporal Knowledge Graph	Bitemporal Bayesian knowledge graph
Conflict Engine	Intelligent conflict resolution (hallucination defense)
Semantic Retrieval	Curiosity-driven semantic retrieval

Adaptive Scorer	Multi-strategy adaptive scoring subsystem
Cognitive RAG	Advanced cognitive RAG pipeline
Integration Hub	Extended tools for external AI agent synergy

D. Pricing Reference Table

Tier	USD (Monthly)	USD (Yearly)	INR (Monthly)	INR (Yearly)
Free	$0	$0	₹0	₹0
Starter	$9/mo	$7/mo	₹749/mo	₹599/mo
Pro	$19/mo	$15/mo	₹1,499/mo	₹1,199/mo
Team	$15/user/mo	$12/user/mo	₹1,249/user/mo	₹999/user/mo
Enterprise	Custom	Custom	Custom	Custom

E. Document History

Version	Date	Author	Changes
1.0	May 30, 2026	Piyush Kumar	Initial product brief — Asana format
1.1	May 31, 2026	Piyush Kumar	Deep architectural audit: added Ghost Meeting Tutorial, Health Dashboard, Device Management, Battery-Aware Scheduling, Widget Quick Actions, 3-Tier Search, Multi-Mode AI Suggestions. Sanitized appendix metrics and removed proprietary internal codebase metrics. Added Epic 5 user stories. Expanded differentiating features table (+8 rows).
